

Summary:

- 1) 15 yr conventional mortgage rate: We have a separate process for the conventional 15 yr mortgage rates similar to the 30 yr mortgage rate where we compute daily CMM102, a current coupon model for 15 yr (coefficients for 2/5/10/30 will be 25%, 31%, 37% and 7%), weekly primary vs secondary spread and a decay to the long term servicing spread assumption.
- 2) 15 yr Ginnie Mae mortgage rate: It will also be similar to the 30 yr Ginnie Mae Mortgage rate process where we compute the monthly spread between Conventional and Ginnie Mae 15 yr based on the daily 15 yr CMM102 of FN and G2 and a decay to the long term servicing spread assumption of -7 bps.
- 3) 20 yr conventional and Ginnie Mae mortgage rate process: Since we have don't good data for CMM102, we will assume that the 20 yr mortgage rate is linearly interpolated between 30 yr and 15 yr. Based on empirical zero sato rates, we will assume 20 yr = 33.33% of 15 yr and 66.67% of 30 yr for both conventional and Ginnie Mae.
- 4) LLPA, PMI and MIP: There are some differences between 15/20 and 30 yr. For example, conventional 15 yr does not have LLPA adjustment for FICO/LTV but has the same adjustment for investor property and loan size. There is a separate PMI for 15/20 yr and MIP for 15 yr. Notice that point to rate adjustment will also be different: 1pt = 33.33 bps (15 yr) and 1pt = 30 bps (20 yr). We will need to verify that the algorithm for PMI and MIP cancellations still applies to 15/20 yr.
- 5) Incentive: For 15 yr conventional and Ginnie Mae, the incentive calculation will be similar to 30 yr once we use the corresponding 15 yr rates and LLPA/PMI/MIP. For 20 yr, it will be a function of four incentives (FN20->FN20, FN20->FN15, FN20->GN20, FN20->GN15) and a linear interpolation between refi into 20 vs 15 as a function of wam (100% 20->20 and 0% 20->15 when wam is 240 and 0% 20->20 and 100% 20->15 when wam is 180).
- 6) CDU: The cdu format for 15 yr models should be similar to 30 yr. For 20 yr, we may need additional incentive fields. The database team will generate the following preliminary spec pool cdu FHL15.250.15(GEN), FHL20.300.16(GEN),GNMII15.250.16(GEN) and GNMII20.350.15(GEN) for analytics.
- 7) Json: For now we will assume the json parameters for 15/20 will be identical to the corresponding 30 yr. One issue that needs to be addressed is how to call the appropriate model for the four sample spec pool listed above.
- 8) Conventional default model: We want to migrate the cdr model for conventional collateral to a structure similar to Ginnie Mae. We will first do that for 15/20 yr models. This means that we will have d0, d30, d60 and d90 for conventional mortgages. Notice that buyout model will always be 100%. We will probably enhance the crr calculations for d30, d60 and d90 by introducing a time dependent multiplier as conventional borrower with d60 or worse delinquency will be less likely to refinance.
- 9) Media effect: We will use the same media as the conventional 30 yr universe